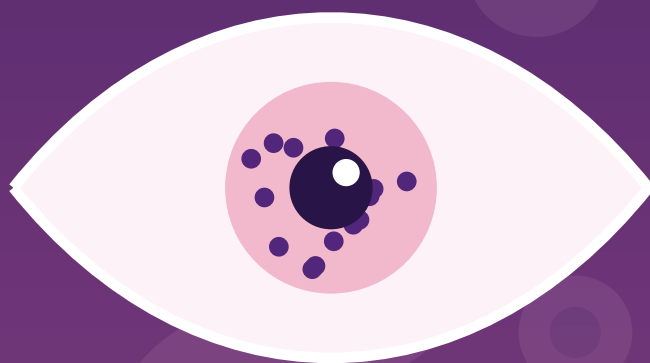




littleEye AI

Manual for physicians and students

version 3.2 · free · iPhone, iPad, Mac

Training and evaluating AI models from histology slides —
entirely on-device, nothing in the cloud.

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forensika.eu

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How to read this manual

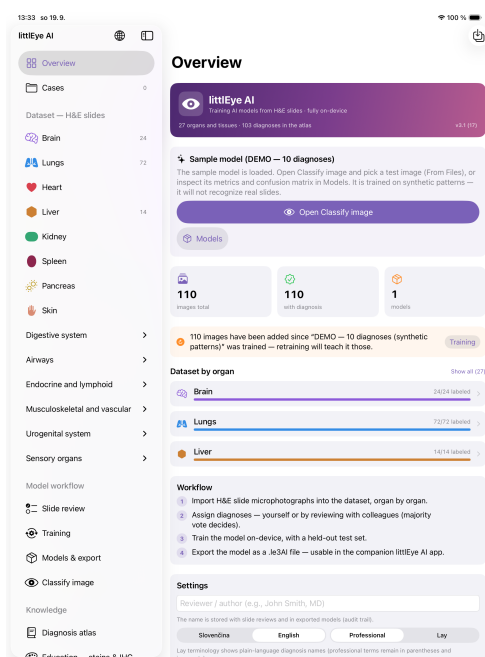
Chapters 2 – 6 are the path from an empty app to your first own model — read them in order. The other chapters are for reference. Screenshots are from iPad; on iPhone the layout is the same, only the sidebar opens with the button at the top left. The Mac has its own chapter 13, the image-analysis tools chapter 14.

1 What littlEye AI is and who it is for

the idea · free · privacy

littlEye AI grew from a simple idea: the microscopic world gives meaning to the macroscopic one. It is built for forensic pathologists and pathologists in practice, and above all for medical students as a companion to histology and pathology.

The app does three things: it collects H&E; microphotographs with diagnoses, trains a small model from them right on your iPad, iPhone or Mac, and evaluates new images — always with a confidence percentage, never without one. On top of that it adds a knowledge base: a diagnosis atlas, an education module on stains and immunohistochemistry, and the list of sources the content is based on.



Overview: dataset status, sample model, workflow and settings in one place.

Free — and why

Since version 3.0 littlEye AI is free and fully functional. No subscription, no feature locked behind a payment, no advertising. The people it was built for — students and junior colleagues — are the same people least able to afford it; and the point of the project is that departments exchange trained models. That needs people, not revenue. Those who wish can make a voluntary contribution (chapter 14). It unlocks nothing.

Privacy as a principle

The entire app works on-device. Images, cases and models never leave it, and the app makes no network connections of its own — the only exceptions are the voluntary contribution through the App Store and literature links, which open in the browser. For forensic medicine this is not an extra feature but the condition without which the app could not exist.

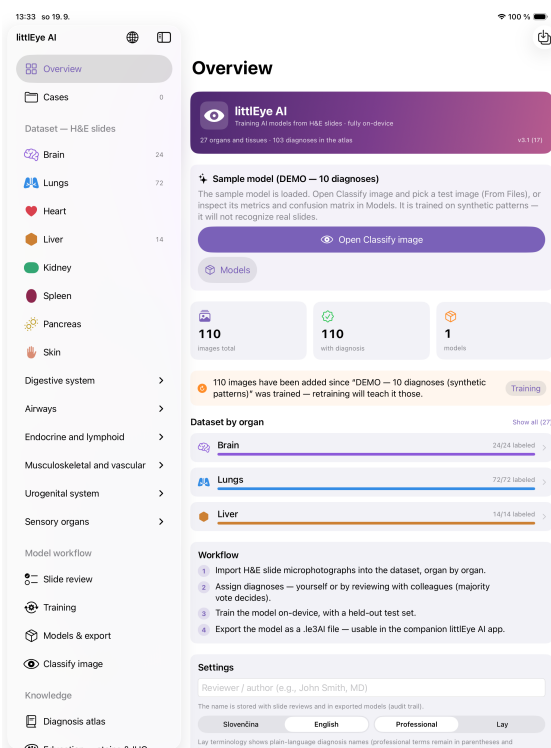
What littlEye AI is not

It is not a medical device and does not give a diagnosis. It is a research and decision-support tool: the model suggests, the physician decides. That sentence is on every content screen of the app on purpose.

2 First launch

language, name, sample model

- 1 On first launch the welcome screen appears. Choose the language (Slovak / English) — it can be switched any time with the globe in the sidebar.
- 2 Enter the reviewer name. It is stored with every slide review and in every exported model — an audit trail, not a registration. It is not sent anywhere.
- 3 On the Overview tap Load sample model. DEMO has 10 diagnoses trained on synthetic patterns — not real slides — and lets you try evaluation, metrics and the confusion matrix right away.
- 4 The Professional / Lay terminology switch (Settings on the Overview) shows diagnoses either professionally (“steatosis”) or plainly (“fatty liver”). Exports and training always use professional keys, so the switch breaks nothing.



Tip for departments

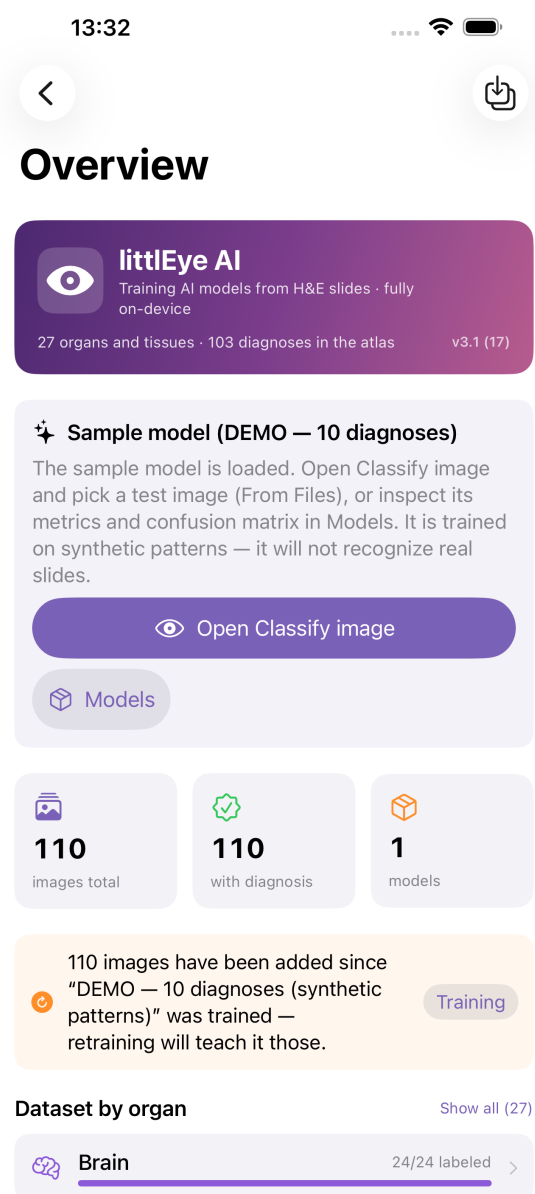
Set a real reviewer name on every device — in colleague review the majority of reviews decides, and the exported model will show who trained it.

3 Dataset: images and diagnoses

import, drag & drop, magnification, stain, crop

The dataset is organised by organ (27 organs and tissues in seven systems). Each image has an organ, optionally a diagnosis, magnification and stain. The model learns from the whole field of view — so the finding should fill a substantial part of the image.

- 1 Pick an organ in the sidebar. At the top right set the import magnification and stain — they are recorded with every image and the model remembers them later.
- 2 Import images from Photos or Files (JPEG, PNG, TIFF, HEIC). Larger batches import in the background with “N / M” in the toolbar. Large scans are normalised to 2048 px.
- 3 On iPad and Mac you can also drag images straight into the organ window — from Photos, Files or Finder.
- 4 Tap an image to open its details: zoom, diagnosis, note, reviews. On the Mac the details open in a separate window.
- 5 If an image contains several findings, use a crop: in the details drag to mark the finding and save it as a separate training image. That is how you tell the model where to look.



The most common trap: the model learns the stain, not the tissue

If a whole class comes from one slide, the model does not learn the tissue but the tint of that slide — and fails on a slide from another laboratory. Every class should have images from at least three different cases or blocks, ideally from different staining runs too. Since 3.1 Training shows you this with a colour traffic light (chapter 5).

Capture rules

- One magnification per model (100× or 200× recommended) — do not mix.

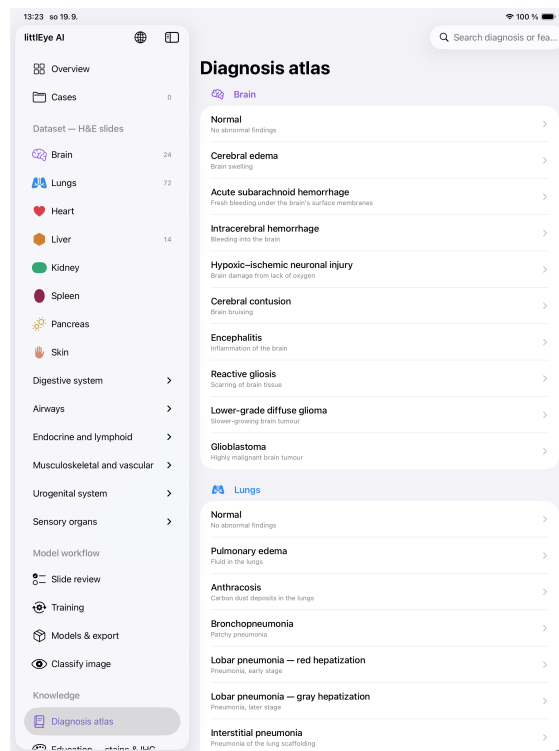
- Sharp, evenly exposed images with correct white balance.
- Do not photograph section edges, folds, creases, bubbles or staining artefacts.
- Several images per slide from different representative places; the finding should fill most of the field of view.
- “Normal” is a full class — the model needs it as much as the pathologies.

4 Slide review by colleagues

audit trail, majority decides

Anyone can assign a diagnosis directly in the image details. For joint work, review is better: each colleague goes through the images under their own name and the final diagnosis is set by the majority of reviews. Every review is stored with name and date.

- 1 Open Slide review, choose an organ and start a session. Images without a diagnosis come first.
- 2 For each image pick a diagnosis from the list. Long-press a diagnosis to open its atlas description — this unifies criteria across the team.
- 3 If unsure, skip the image. A wrong label hurts more than a missing one.
- 4 A manual assignment in the details overrides the majority — use it for the final decision when reviews disagree.



Example from practice

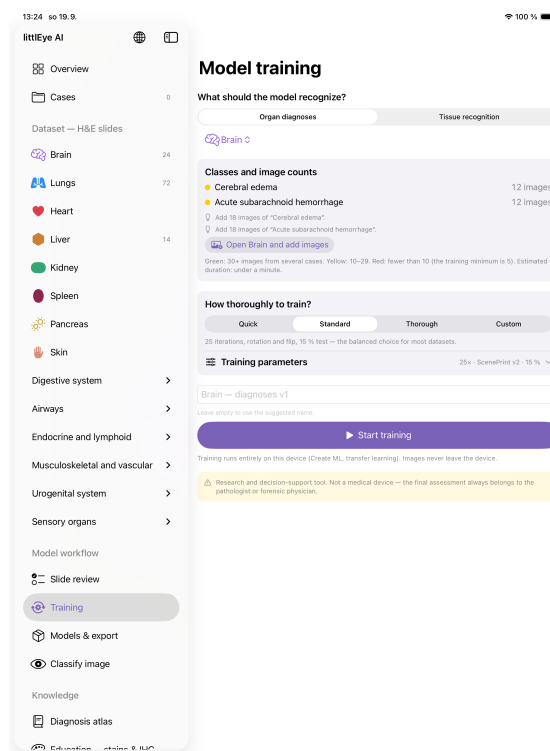
Three colleagues review 40 lung images. They disagree on six (oedema vs. congestion). The majority sets the diagnosis; the team looks at two contested images together and decides manually in the details. Only then do they go into training.

5 Training a model

readiness traffic light, profiles, time estimate

Training runs entirely on-device (Create ML, transfer learning over the ScenePrint extractor). Version 3.1 is built so that you need not know terms like holdout or iteration: the app tells you whether the dataset is ready and offers three profiles.

- 1 Choose the type: Organ diagnoses (a single-organ model) or Tissue recognition (a model that tells organs apart). Organs are split into “ready” and “not enough images yet”.
- 2 The Classes and image counts card shows a traffic light: green 30+ images from several cases, yellow 10–29, red fewer than 10. Below it is a concrete hint — “Add 12 images of Normal — ideally from another case” — and a button straight into the organ.
- 3 Pick a profile: Quick (in a minute you know whether the data works), Standard (for most datasets) or Thorough (the most honest estimate, takes longest). Advanced parameters stay below for those who want them.
- 4 The model name is prefilled (“Lungs — diagnoses v2”) — the version number grows by itself. Press Start training. The duration estimate is under the traffic light; training can be cancelled any time.



What happens under the hood

A portion of images (10–20 % by profile) is set aside as an independent test set — the model never sees them during training and accuracy on them is the most honest estimate of performance. If the smallest class has fewer than 20 images after the holdout, the app also disables Create ML's internal validation: with 1–2 images it would show 0 % or 100 % and mean nothing.

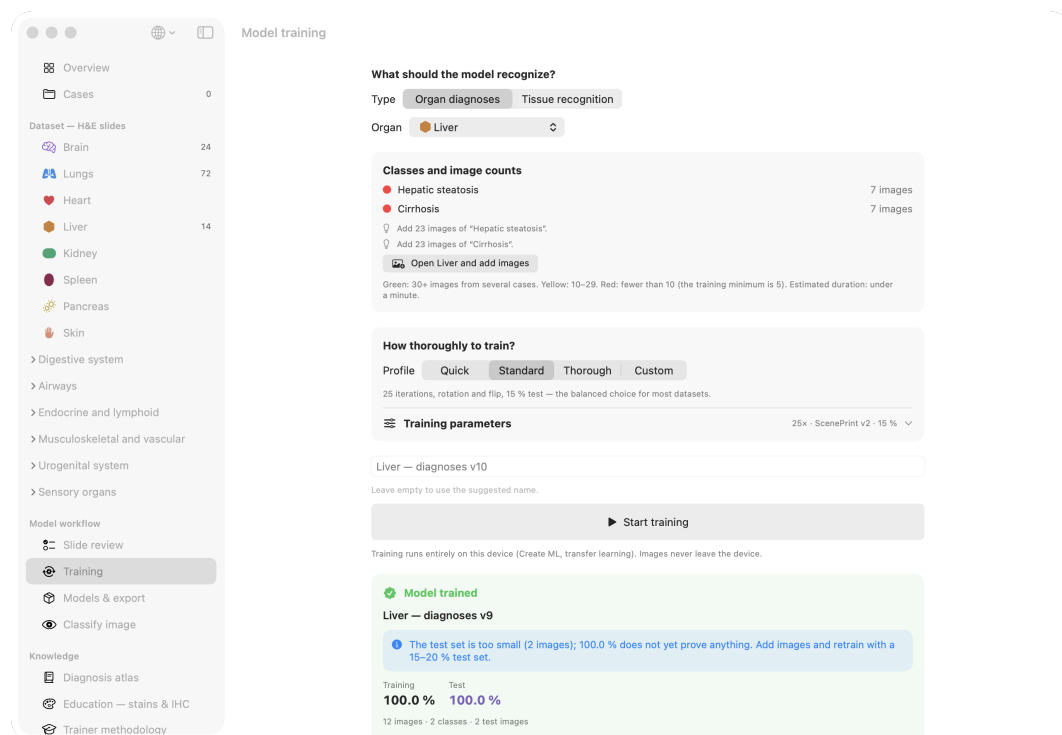
Minimum and recommendation

The minimum is 2 classes of 5 images — enough to try the path, not to make a model. A reasonable start is 30+ images per class from several cases, balanced between classes. A model with 112 normal images and 3 infarct images will answer “normal” to everything.

6 The training result and what to do with it

quality sentence, confusion matrix, what to add

After training you get a card with one sentence about quality, the metrics and the confusion matrix. The sentence is derived from the numbers by rules — no further AI — and says what to do with the model.



The result card on the Mac: quality sentence, training / validation / test, confusion matrix and the buttons Try on an image, What to add, Save .le3AI.

What the sentences mean

- Usable for triage (test ≥ 90 %) — the model can pre-sort images; the final assessment remains with the physician.
- Indicative (75–90 %) — good as a second opinion, not for triage.
- Unsure about class X — confuses it with Y — the confusion matrix showed a specific weakness; the button What to add to improve it lists which classes need images.
- Overfitted (training 20+ % better than test) — the model memorised images instead of tissue. Variety (other cases) helps, not more iterations.
- The test set is too small — with a handful of test images the number proves nothing. Add images and retrain.

The confusion matrix

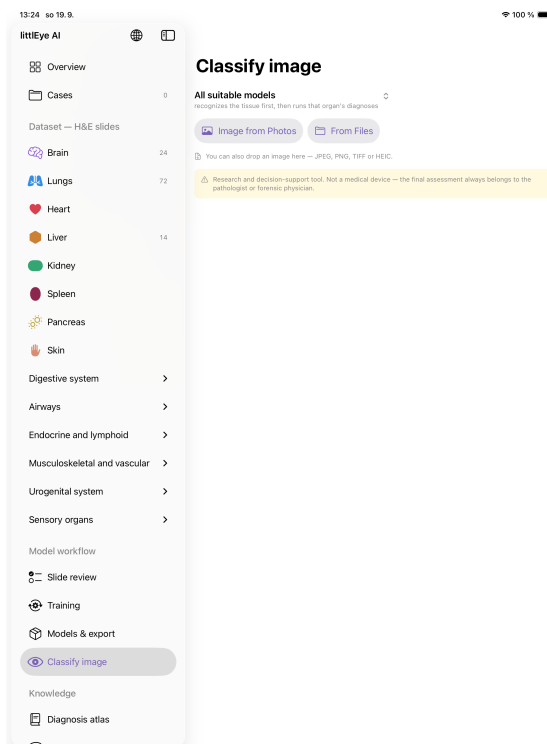
Row = actual diagnosis, column = what the model said. Numbers on the diagonal are correct answers. If the “Oedema” row has many in the “Congestion” column, the model confuses those two — and exactly those two classes need more varied images. Below the matrix is each class's sensitivity and precision.

The model is stored inside the app — you will find it in Models & export. A .le3AI file is created only when you export (chapter 10). The button Try on an image takes you straight to evaluation with this model.

7 Evaluating an image

model chaining, field-of-view crop, preliminary diagnosis

- 1 Open Classify image. The default All suitable models first recognises the tissue and then runs only the diagnostic models of that organ; models of other organs are deliberately skipped and the app says why.
- 2 Add an image from Photos, from Files, or drop it into the window (iPad, Mac). Above the image you see what the model was trained on (“100× · H&E”) — the key fact when tuning inaccurate results.
- 3 For an overview image use Select field of view (crop) and mark a patch the size of one field of view. Only the crop is evaluated — structures come to the scale the model was trained on. This alone fixes most nonsensical results.
- 4 The result always carries a confidence percentage. For the most confident class the app offers a preliminary diagnosis: microscopic description and a suggested conclusion from the knowledge base, with a Copy report draft button and source citations.



How to get the most accurate result from a poor image

The model does not compare the image with a textbook — it compares it with what it saw in training. Match the scale with a crop, remove from the field everything that is not the finding, evaluate 3–5 different fields from the same slide: a consistent answer is trustworthy, scattered answers mean the image is outside what the model knows. And check the class list — the model only ever chooses among the classes it learned.

8 The learning loop

learn from a correction, when to retrain

Core ML models cannot be fine-tuned — they are “baked”. Learning therefore goes through the dataset: you save the corrected image with the right diagnosis, and at the next retraining a new model is created that already knows it.

- In Classify image, below the result, is Teach from this image: pick the correct organ and diagnosis (the recognised organ is prefilled) and the image is saved into the dataset.
- The app counts how many labelled images were added since each model was trained. At 10 or more, the Overview and Models & export show the prompt “N images added since training — retrain”.
- Retraining creates a model with the next version number; the old one stays until you delete it — so you can compare whether the new one really helped.

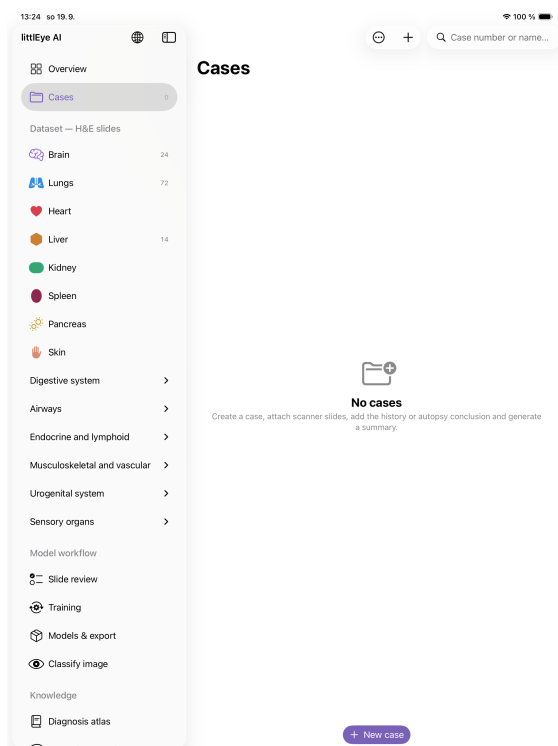
Example

A liver model labels steatosis as normal. In the Teach section the physician picks “Steatosis” and saves the image. After ten such corrections the app suggests retraining; the new v3 model has 15 % higher sensitivity for steatosis — visible in the confusion matrix.

9 Cases

slides per case, assessment, backup

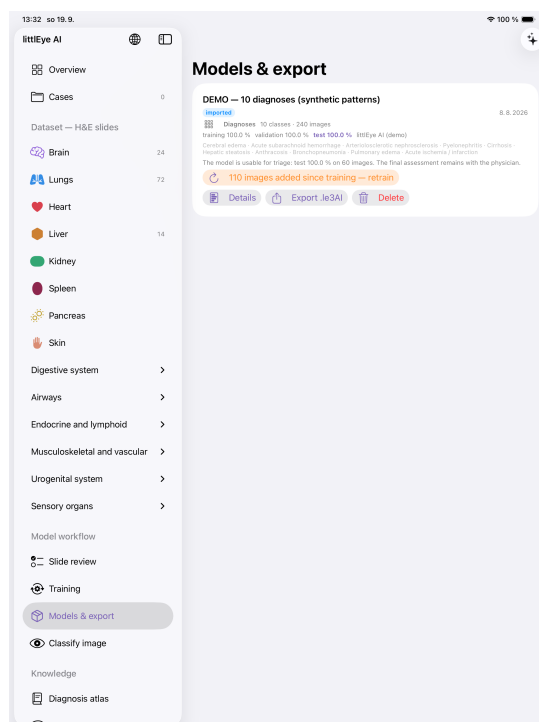
- 1 Cases (top of the sidebar) group scanner slides under a protocol number. Case slides with a diagnosis also count towards the training dataset.
- 2 Add context to the case: history, autopsy conclusion, toxicology. They are inputs for the assessment and are not sent anywhere.
- 3 Assess case evaluates all slides (physician diagnoses + available models) and generates a concise summary: findings by organ, a “slides only” and a “with case data” conclusion. The Copy button pastes it into Word.
- 4 If you have a tissue-recognition model, every slide is checked with it: on a mismatch with $\geq 85\%$ confidence the summary gets an Attention — tissue check section (suspected slide mix-up).
- 5 Cases →  → Back up archive to disk writes a complete copy (images, models, cases) to an external drive, USB-C or SMB. A backup is always complete, never incremental.



10 Sharing: .le3AI models and .le3set datasets

what travels, what does not, encryption

Collective work rests on two files. .le3AI is a model: one file with the model, metrics, confusion matrix, magnification, stain, data provenance and licence. .le3set is a dataset: images with diagnoses and reviews — without any case or patient data.



Models & export: every model with metrics, the quality sentence and the buttons Details, Export .le3AI, Delete.

Why images are shared, not just models

A model cannot be extended with more images. If a department wants to combine material from three hospitals, each sends its .le3set, the recipient merges them (re-importing the same material does not duplicate it) and retrain the model from scratch on the larger whole. The finished model then goes back as .le3AI.

- Export: Models & export → Export .le3AI; dataset: organ → toolbar → Sharing → Share slides (.le3set) (respects the diagnosis filter).
- Import: double-click the file, AirDrop, Files, or  on the Mac / iPad with a keyboard. Images from a .le3set sort themselves into organs.
- Both files are encrypted (AES-GCM) — only littlEye AI opens them and a changed byte is detected. A .le3AI carrying a model in .mlpackage format (chapter 11) requires version 3.1 or later.

Data provenance is part of the model

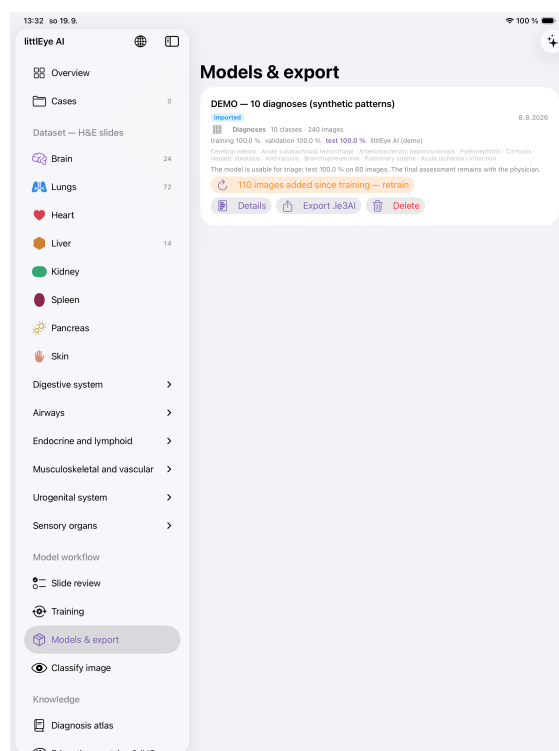
A model travels for years. Without a recorded source and licence nobody can later judge honestly what may be done with it — whether commercial use is allowed, whether the licence must be preserved, whom to cite. That is why .le3AI carries them and why the app requires them when importing a foreign model.

11 Importing a model trained outside the app

.mlpackage / .mlmodel, provenance, in-app test

Since version 3.1 littlEye AI also accepts a model trained outside the app — typically in Python (PyTorch) and converted with coremltools to `.mlpackage` or `.mlmodel`. Everything the app stands on is preserved: the model runs on-device, its provenance is recorded and the app measures the metrics itself.

- 1 Models & export →  → Import external Core ML model (on the Mac also File → O). Choose a `.mlpackage` or `.mlmodel`.
- 2 The app checks the model: it must have one image input and classifier outputs (class + probabilities). Otherwise it is rejected with a precise hint what to change in the conversion.
- 3 Map each model class to a catalogue diagnosis (prefilled when the class name matches) or type your own name. This makes the preliminary diagnosis and the in-app test work.
- 4 Fill in the data source and licence — required; without them the import does not complete.
- 5 After import press Test on my N images: the model runs on images from your dataset with the same diagnoses and the result (accuracy, confusion matrix) is recorded as its only metric. The app never shows numbers taken over from elsewhere.



Conversion in Python (minimum)

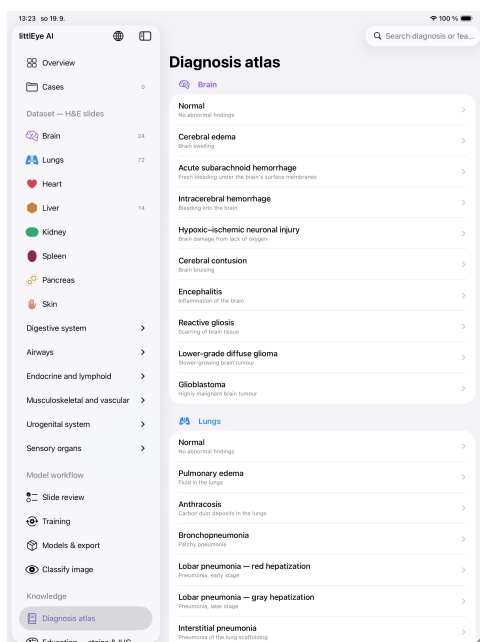
```
import coremltools as ct
mlmodel = ct.convert(traced_model,
inputs=[ct.ImageType(name="image", shape=(1, 3, 224, 224), scale=1/255.0)],
classifier_config=ct.ClassifierConfig(["pluca.edem", "pluca.antrakoza"]),
minimum_deployment_target=ct.target.iOS17)
mlmodel.save("model.mlpackage")
```

If you use littlEye AI catalogue IDs as class names (e.g. `pluca.edem`), the mapping fills itself in at import. Models over 200 MB are hard to send by mail; over 1 GB the app rejects them.

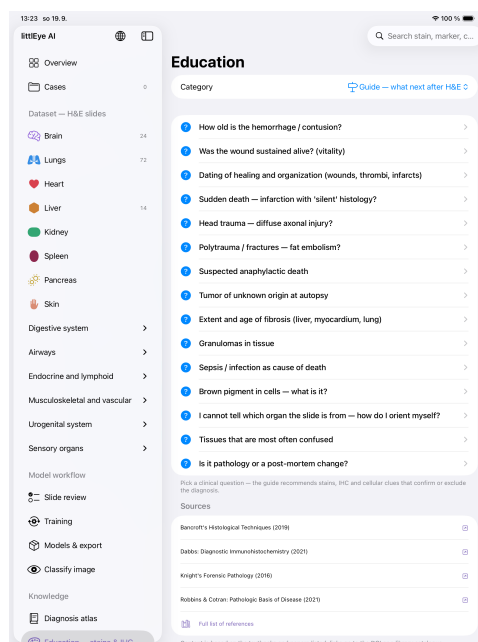
12

Atlas, Education, Sources

103 diagnoses, stains, IHC, citations



Diagnosis atlas — 103 diagnoses across 27 organs, search by name or feature.



Education — the “what next after H&E” guide, stains, IHC, cells, pigments.

Diagnosis atlas

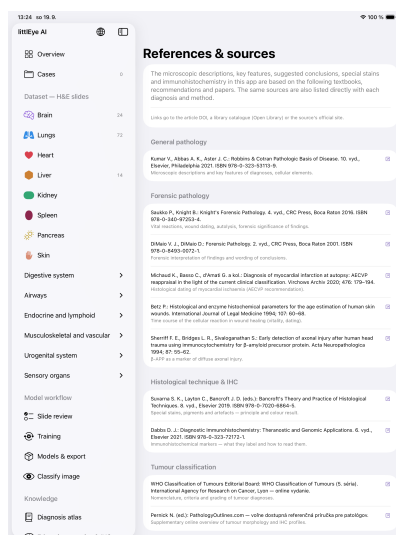
Each diagnosis has a microscopic appearance, key features, a suggested wording of the conclusion (Copy button), recommended next steps (special stains, IHC) and a Sources card with links. The atlas helps reviewers unify criteria and is available directly during review.

Education

The “what next after H&E” guide starts from a clinical question (dating a haemorrhage, wound vitality, silent infarct, DAI, fat embolism, anaphylaxis, tumour of unknown origin, autolysis vs. pathology...) and recommends stains, IHC markers and cellular clues that support or exclude the diagnosis.

References and sources

All medical content is cited — textbooks (Robbins & Cotran, Knight, DiMaio, Bancroft, Dabbs), the AECVP recommendation on myocardial infarction, papers on wound dating and β -APP, the WHO classification, PathologyOutlines and the NCT-CRC dataset. Links go to the DOI, a library catalogue or the official site. You find them with each item and collected under Knowledge → References & sources.

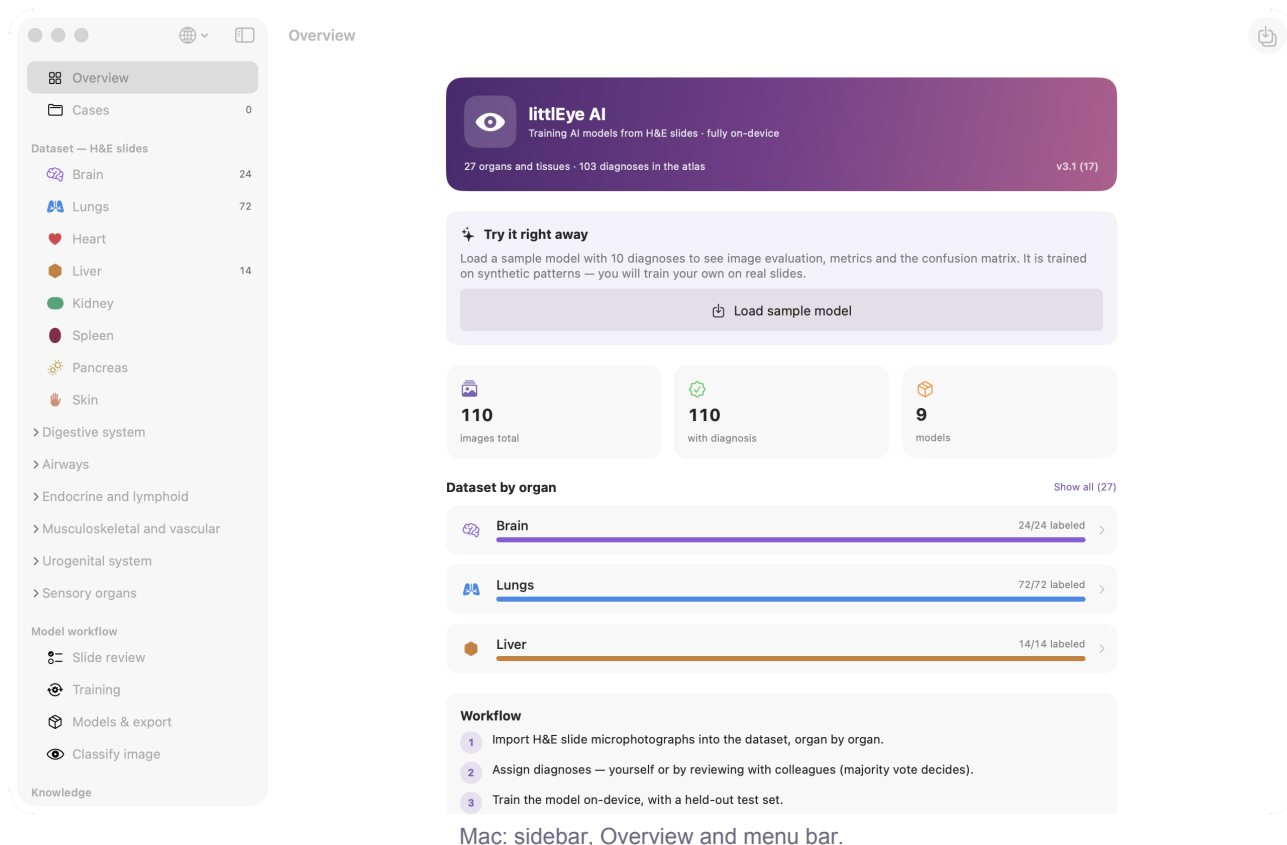


References & sources — every source is a link.

13 littlEye AI on the Mac

menus, shortcuts, windows, drag & drop

On an Apple silicon Mac littlEye AI is a native app with a menu bar, keyboard shortcuts and multiple windows. Training on the Mac uses the whole chip and is fastest — the Mac is therefore the department's ideal “training machine”, while iPad and iPhone collect and review.



Mac: sidebar, Overview and menu bar.

Shortcuts (also work on iPad with a keyboard)

⌘ Import model or dataset

⌘⌘ Import external Core ML model

⌘1 ... ⌘5 Overview · Cases · Classify · Training · Models

⌘6 ... ⌘0 Atlas · Education · Methodology · Sources · About

⌘, Settings (name, language, terminology, icon)

Esc Closes the open sheet

Windows and drag & drop

- Slide details open in a separate window (also right-click → Open in a separate window) — you can keep it next to the case.
- A .le3AI or .le3set dropped onto the window is imported; images dragged into an organ or into Classify image are loaded.
- Training progress also shows as a percentage on the Dock icon.
- Help contains links to the guides (PDF), the website and Support the project.

Microscope images on the Mac

The app needs no drivers: microscope camera → export → Files/iCloud/AirDrop; USB-C UVC cameras work natively; slide scanners with TIFF/JPEG export. Details are in the app under Hardware & capture.

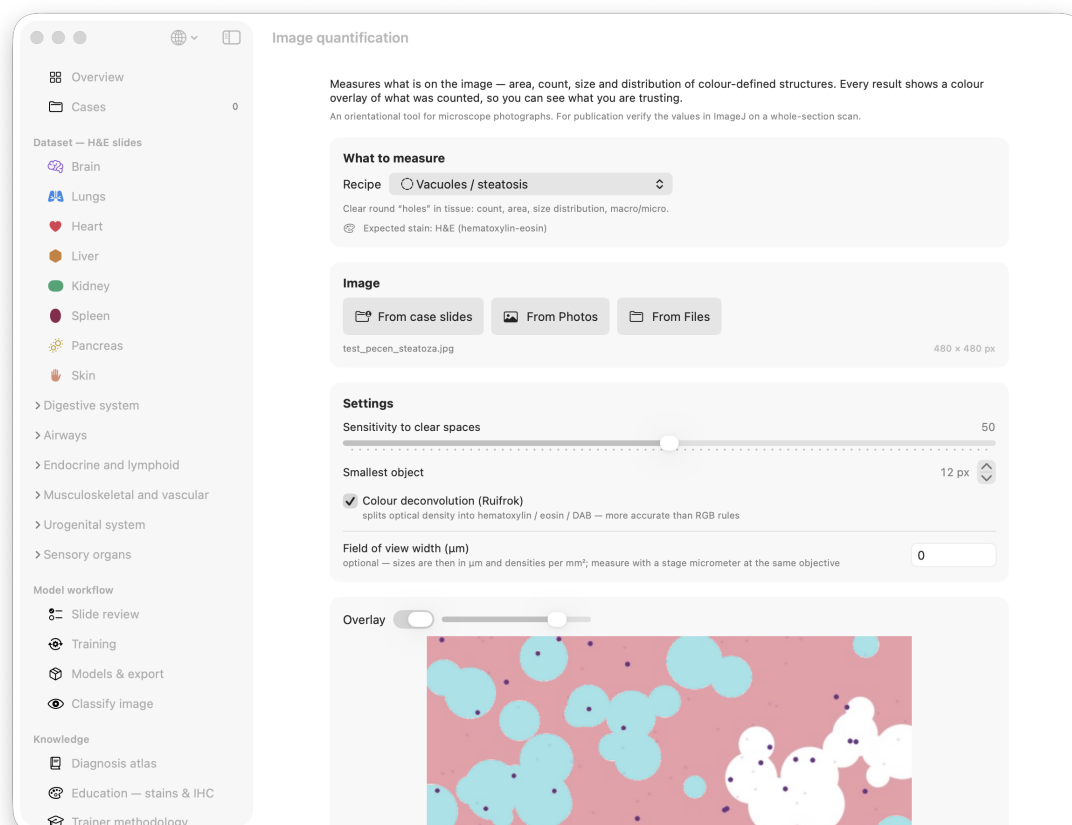
14 Image quantification and desmin IHC

lipofuscin, vacuoles, erythrocytes, nuclei · CPA and indices

Since version 3.2 littlEye AI has two tools that use no trained model but measure colour and shape on the image directly: Image quantification (area, count, size and distribution of colour-defined structures) and Desmin IHC (a calculator of CPA and indices based on the author's dissertation). Both sit in the sidebar under Tools — image analysis; on the Mac also in the Go menu (Shift-Command-Q, Shift-Command-D).

Image quantification

Pick a recipe (what to measure), load an image — from case slides, Photos, Files or by dropping it — and the result is immediate: a number, a colour overlay of what was just counted, and a sentence on what the number means and what it does not. The sensitivity slider and the overlay exist so that you always see what you are trusting.



The Vacuoles / steatosis recipe on a synthetic liver image: turquoise = counted vacuoles, white = excluded (a large irregular area at the border).

Recipes

- Lipofuscin (H&E) — area, count and size of yellow-brown granules, granules per 100 nuclei (a stand-in for “per cell”) and the perinuclear share. Colour alone does not separate lipofuscin from hemosiderin or formalin pigment — confirm with Perls, PAS, long Ziehl-Neelsen.
- Vacuoles / steatosis (H&E) — clear round spaces: area, count, macro- vs. microvesicular by nuclear diameter, size histogram; NASH CRN bands as orientation only (they refer to the share of hepatocytes, not to area).

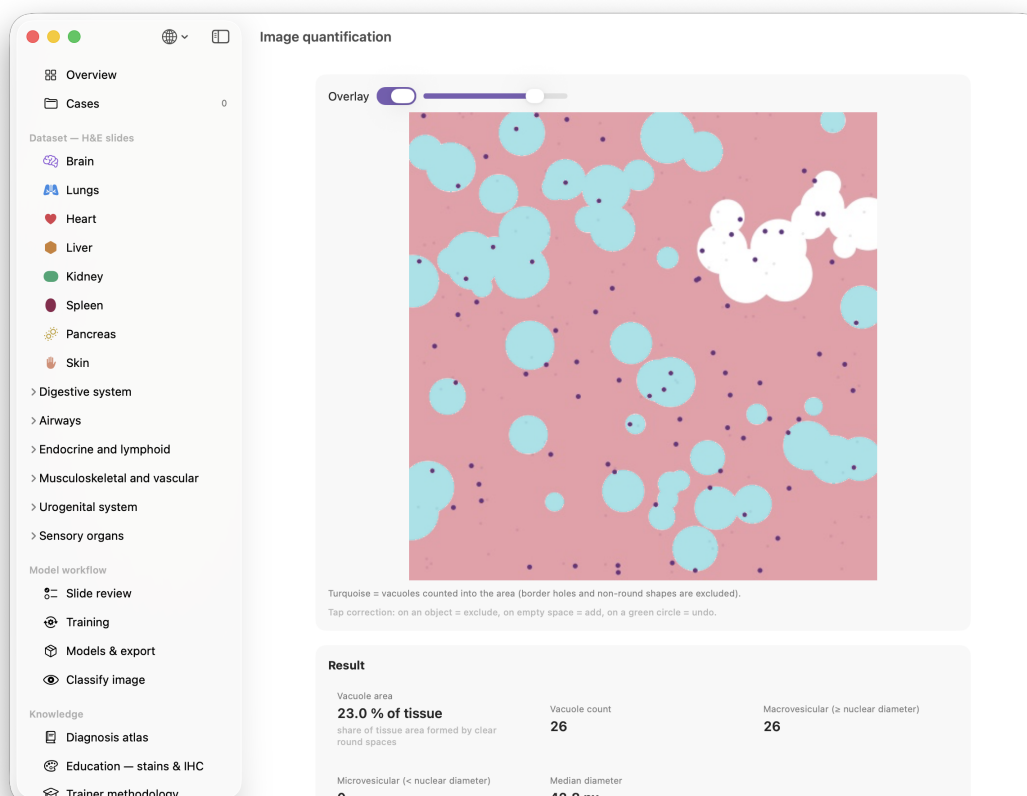
- Erythrocytes (H&E) — area of bright red, extent across the field (diffuse vs. focal), the largest contiguous focus. Area does not separate congestion from hemorrhage — that is for the physician by location.
- Nuclei and PMNL (H&E) — count and density of hematoxylin nuclei; small dark lobulated ones as an orientational PMNL share. Tap correction: on an object = exclude, on empty space = add.
- Desmin — CPA (DAB) and Myocytes on H&E — the image helper for the Desmin IHC calculator: CPA H-DAB (strict threshold), more-brown (wide threshold) and the pink myocyte area; Use in the desmin calculator carries them over as a new section.

Scale and units

Without a scale, sizes are in pixels and densities per field. Enter the field width in μm (measure with a stage micrometer at the same objective) and you get μm , μm^2 and counts per mm^2 . The value is remembered.

What quantification is not

It measures colour and shape on a photograph — not a diagnosis. Thresholds are sensitive to exposure, white balance and section thickness; only images from the same protocol are comparable. For publication verify the values in ImageJ on a whole-section scan. Copy summary puts all numbers and settings on the clipboard.



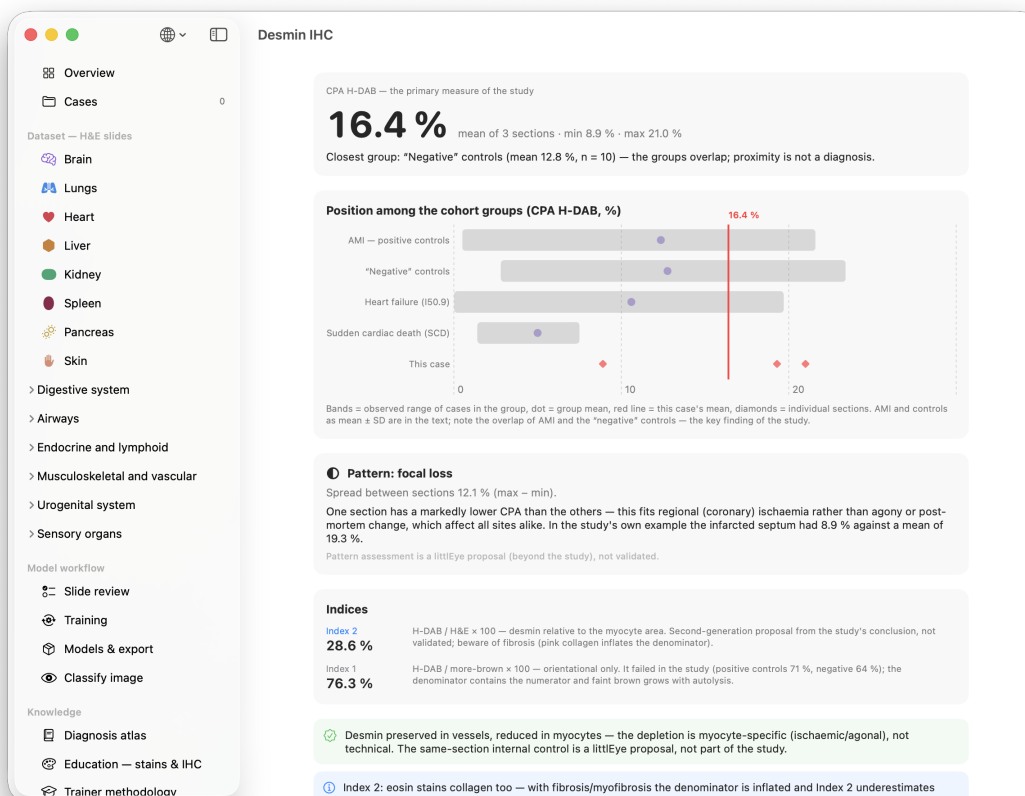
The result: every number carries a note on what it means; below it the size histogram and the recipe's caveats.

Desmin IHC — CPA and indices

Desmin is a structural protein of cardiomyocytes that is lost in ischaemia and in agonal hypoxia alike — the loss of staining is the finding. The author's dissertation (Šikuta 2021, $n = 89$) quantified the extent of this loss as CPA (Colour Proportionate Area, % of area with brown DAB) and found that the CPA in confirmed

infarction (12.4 ± 7.9 %) and in “negative” controls (12.8 ± 5.4 %) overlapped: depletion accompanies any death with an agonal phase. The calculator therefore gives no verdict.

- 1 Enter each sampling site separately (septum, anterior and posterior LV wall...) with its CPA H-DAB; more-brown and H&E are optional for the indices. Infarction is focal — a mean dilutes it.
- 2 Fill in the confounders: the death → cooling interval (above 12 h the CPA falls by itself, markedly from 32 h), pancreatic autolysis, resuscitation.
- 3 Check desmin in the vascular smooth muscle of the same section — the cheapest control: preserved in vessels + lost in myocytes = the loss is myocyte-specific; lost in both = autolysis or a staining failure.
- 4 Evaluate: you get the mean, minimum and maximum, the pattern (focal loss / diffusely low / even), a chart of the position among the four cohort groups, the indices and a list of what could have lowered the number without ischaemia.



The result on the study's own example (infarcted septum 8.9 % vs. 19.3 and 21.0 %): focal-loss pattern, position among the groups, Index 2 and 1 with their caveats.

Indices

- Index 2 = $\text{H-DAB} / \text{H\&E} \times 100$ — desmin relative to the pink myocyte area on a serial section. A second-generation proposal from the study's conclusion, not validated; eosin stains collagen too, so with fibrosis the index underestimates desmin.
- Index 3 = $\text{heart} / \text{skeletal muscle of the same body} \times 100$ — a littleEye proposal: the denominator shares the post-mortem interval, fixation and staining batch, but not the agonal ischaemia of the myocardium. Not validated.
- Index 1 = $\text{H-DAB} / \text{more-brown} \times 100$ — failed in the study (positive controls 71 %, negative 64 %); kept only as an orientational note.

How to read the result

There is no deciding threshold. Informative is a focal loss with sharp demarcation in a well-preserved body; diffuse loss is non-specific (agony, resuscitation, asphyxia, post-mortem interval, putrefaction). The cohort numbers are descriptive data from one institution and protocol, not reference bands. To confirm early ischaemia, C5b-9(m), fibronectin and fibrinogen are better established — they accumulate in ischaemia rather than disappear. The final assessment belongs to the physician.

Where the CPA comes from

The reference path is a whole-section scan and ImageJ with the IHC Toolbox plugin (H-DAB and H-DAB more brown models). The image helper in Quantification gives an orientational value from a microscope photograph — good for triage and teaching, not for publication.

15 Privacy, limits, collaboration and support

what the app does not do · collaboration · voluntary contribution

What the app does not do

- Does not connect to the network (exceptions: the App Store for a contribution, literature links in the browser). No accounts, no telemetry.
- Does not give a diagnosis. The model output is always only a suggestion with a confidence percentage.
- Contains no real slides and no ready-made diagnostic models — only the synthetic DEMO is bundled. You train models yourself or exchange them as files.
- Does not fine-tune models — it retrains them from scratch on a larger dataset (chapter 8).

Limits worth knowing

- The model only ever chooses among the classes it learned. A model of normal tissue will not recognise pneumonia — it labels it as the closest thing it knows.
- If normal and finding come from different image sources, the model reads the scanner's look instead of the tissue; a negative result then excludes nothing. Collect both classes from the same source.
- Before using a model in practice: validation on an independent set, sensitivity/specificity, agreement with a pathologist.

Collaboration

littlEye AI comes from a single institution. For the models and quantification methods to mean anything, they need validation on someone else's material — another scanner, another laboratory, other hands. We are looking for institutions that can offer H&E slides with a clear licence to share, validation of models and recipes on their own images, or co-authorship of methodological publications (desmin CPA, image analysis). We offer co-authorship, credit in the app and models free of charge. The offer appears twice on the Overview and permanently in Settings and About; write to forensika.eu@icloud.com.

Supporting the project

littlEye AI is developed and maintained by one physician in his own time. If it helps you, you can contribute under About → Support the project — a coffee, a lunch, whatever suits, repeatedly if you wish. The App Store does not allow arbitrary amounts, hence fixed levels. Contributions fund upkeep and further development and unlock nothing. The reminder on the Overview goes quiet after a contribution.

Contact and links

Website and guides: dzanino.github.io/littleEyeAI · Models on request and questions: forensika.eu@icloud.com · The Forensika project: forensika.eu

littlEye AI 3.2 (18) · September 2026 · Ján Šikuta, MD, PhD, Institute of Forensic Medicine, Comenius University Bratislava. Developed in collaboration by specialists from three institutes of the Faculty of Medicine.